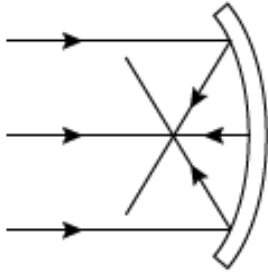


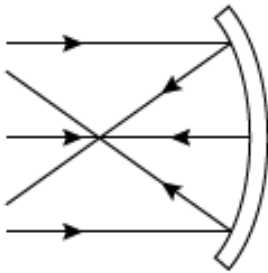
Which of the following spherical mirrors produces a virtual image from the given light rays and has the larger radius of curvature? (Note: Assume that the distance between the object and mirror is greater than the focal length.)

☐ A.



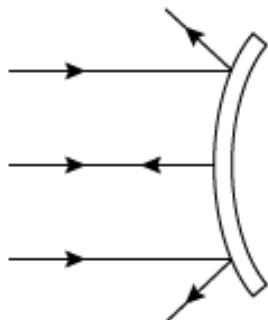
[9%]

☐ B.



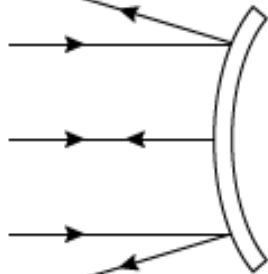
[23%]

☐ C.



[29%]

✓ ☒ D.

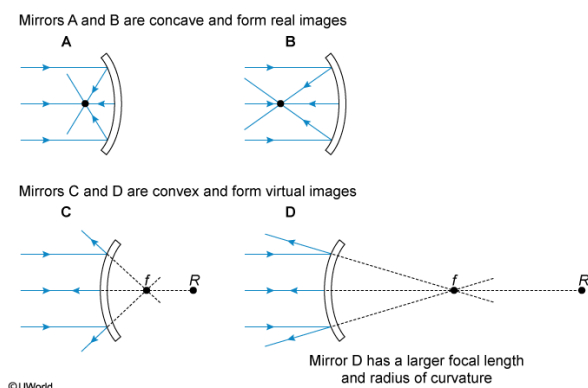


[38%]

Correct

38% answered correctly

Explanation:



Spherical mirrors are shaped like a section of a sphere's surface, and the degree to which they bend light is determined by the mirror's focal length. The principal axis of a mirror is the line that passes through its center and is perpendicular to its surface. A mirror's **focal length** is the distance along the principal axis to the focal point, the location where incident rays parallel to the principal axis converge to or diverge from.

- Concave (converging) mirrors have their reflective surface curved inward, and they converge light rays to form real images in front of the mirror when the distance of the object is greater than the focal length.
- **Convex** (diverging) mirrors have their reflective surface curved outward, and they diverge light rays to form **virtual images** behind the mirror.

The **radius of curvature** R of a spherical mirror is twice the focal length f :

$$R = 2f$$

Of the given options, only mirrors C and D are convex and create virtual images. Because the light rays in mirror D are reflected away at a smaller angle, these rays can be **traced back** behind the mirror to a more distant focal point than that traced by the light rays in mirror C (**Choice C**). Because the radius of curvature is twice the focal length, mirror D also has a larger radius of curvature than mirror C.

(Choices A and B) Although mirror B has a larger radius of curvature, both mirrors A and B are concave and therefore create real, not virtual, images under these conditions.

Educational objective:

Concave (converging) mirrors create real images when the object is placed outside the mirror's focal length, whereas convex (diverging) mirrors always create virtual images. For spherical mirrors, the radius of curvature is twice the focal length: $f = \frac{R}{2}$.